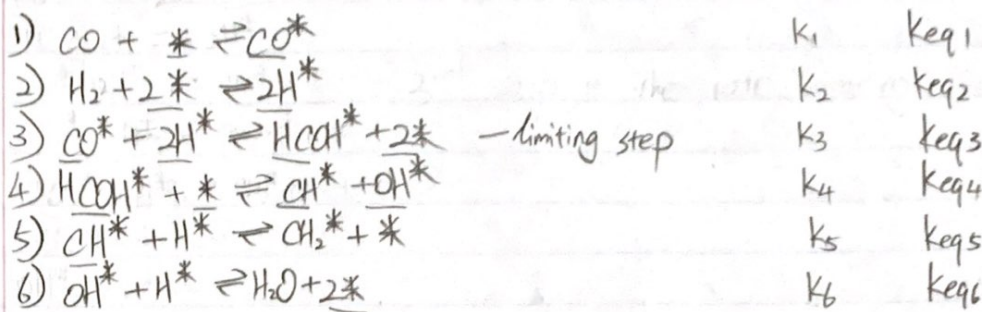
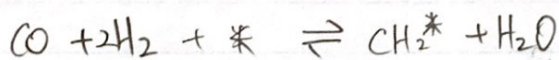
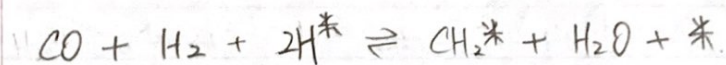




overall:



$$r_1 = -k_1([\text{CO}] - k_{eq1}^{-1}[\text{CO}^*])$$

$$r_2 = -k_2([\text{H}_2] - k_{eq2}^{-1}[\text{H}^*]^2)$$

$$r_3 = -k_3([\text{CO}^*][\text{H}^*]^2 - k_{eq3}^{-1}[\text{HCOH}^*])$$

$$r_4 = -k_4([\text{HCOH}^*] - k_{eq4}^{-1}[\text{CH}^*][\text{OH}^*])$$

$$r_5 = -k_5([\text{CH}^*][\text{H}^*] - k_{eq5}^{-1}[\text{CH}_2^*])$$

$$r_6 = -k_6([\text{OH}^*][\text{H}^*] - k_{eq6}^{-1}[\text{H}_2\text{O}])$$

since step 3 is limiting step

$$\frac{r_1}{k_1} = \frac{r_2}{k_2} = \frac{r_4}{k_4} = \frac{r_5}{k_5} = \frac{r_6}{k_6} = 0$$

$$\Rightarrow [\text{CO}] = [\text{CO}^*]k_{eq1}$$

$$[\text{H}_2] = [\text{H}^*]^2k_{eq2}$$

$$[\text{HCOH}^*] = [\text{CH}^*][\text{OH}^*]k_{eq4}$$

$$[\text{CH}^*][\text{H}^*] = [\text{CH}_2^*]k_{eq5}$$

$$[\text{OH}^*][\text{H}^*] = [\text{H}_2\text{O}]k_{eq6}$$

$$[\text{CO}^*] = \frac{[\text{CO}]}{k_{eq1}}$$

$$[\text{H}^*]^2 = \frac{[\text{H}_2]}{k_{eq2}}$$

$$[\text{CH}^*] = \frac{[\text{CH}_2^*]k_{eq5}}{[\text{H}^*]}$$

$$[\text{OH}^*] = \frac{[\text{H}_2\text{O}]k_{eq6}}{[\text{H}^*]}$$

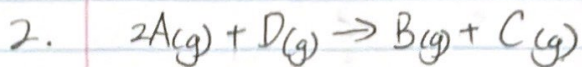
$$r_3 = -k_3 \left(\frac{[\text{CO}]}{k_{eq1}} \cdot \frac{[\text{H}_2]}{k_{eq2}} - k_{eq3} \cdot \left(k_{eq4} \frac{[\text{CH}_2^*]k_{eq5}[\text{H}_2\text{O}]k_{eq6}}{[\text{H}_2]} \right) \right)$$

$$[\text{HCOH}^*] = [\text{CH}^*][\text{OH}^*]$$

$$[\text{HCOH}^*] = k_{eq4} \left(\frac{[\text{CH}_2^*]k_{eq5}[\text{H}_2\text{O}]k_{eq6}}{[\text{H}^*]^2} \right)$$

if $k_{eq_i} = 1$

$$r_3 = -k_3 \left([\text{CO}][\text{H}_2] - \frac{[\text{CH}_2^*][\text{H}_2\text{O}]}{[\text{H}_2]} \right)$$



overall $-r_A = \frac{k_B P_A^2}{(1 + k_A P_A + k_B P_B)^2}$

$r_{ad} = +k_{ad} (P_A C_v - \frac{1}{P_A} C_{A,s})$

$r_{des} = +k_D (C_{B,s} - k_B P_B C_v)$

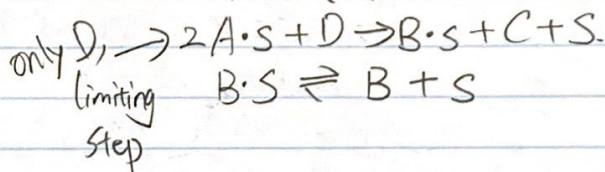
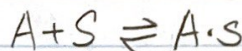
$\frac{r_{ad}}{k_{ad}} = \frac{r_{des}}{k_D} = 0$

$C_{A,s} = k_A P_A C_v$

$C_{B,s} = k_B P_B C_v$

$C_T = C_v + C_{A,s} + C_{B,s}$
 $= C_v + k_A P_A C_v + k_B P_B C_v$

$C_v = \frac{C_T}{(1 + k_A P_A + k_B P_B)}$



reaction step is the limiting step

$-r_A = r_s = k_s (C_{A,s})^2 (P_D) \leftrightarrow \text{non reversible}$

$= k_s \left(k_A P_A \cdot \frac{C_T}{1 + k_A P_A + k_B P_B} \right)^2 (P_D)$

k_s, k_A, C_T are constant $\Rightarrow K$ (a constant)

$-r_A = \frac{K P_A^2 P_D}{(1 + k_A P_A + k_B P_B)^2}$